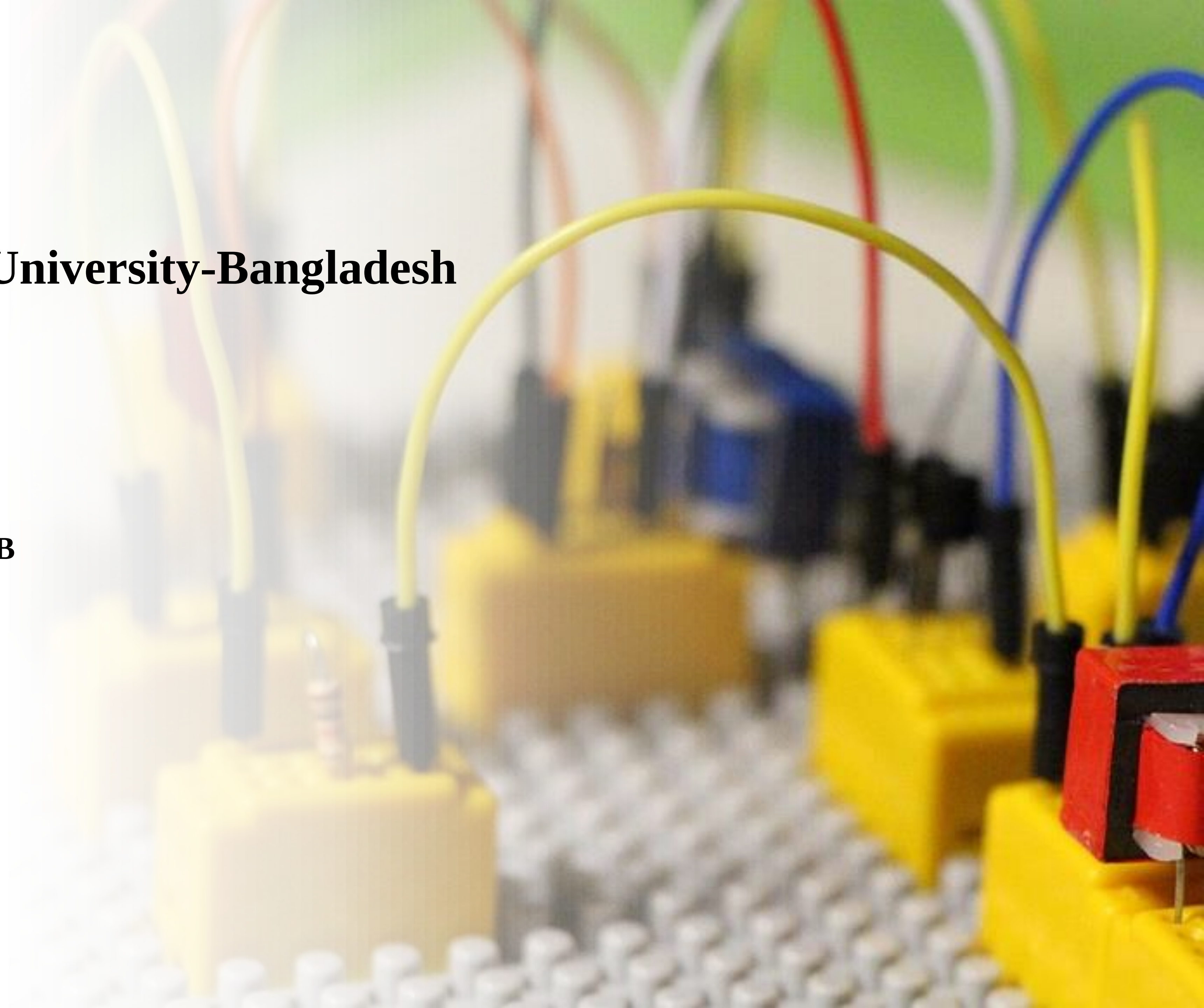
American International University-Bangladesh
Faculty of Science and Technology

Project Presentation
Spring 2024-25

Introduction to Electrical Circuits LAB
Section : O
Group : 1
Faculty : Sadia Rakinur Rab



Project Title:

Laser Security Alarm using LDR and Transistor.

Group Members

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Objectives

This project utilizes

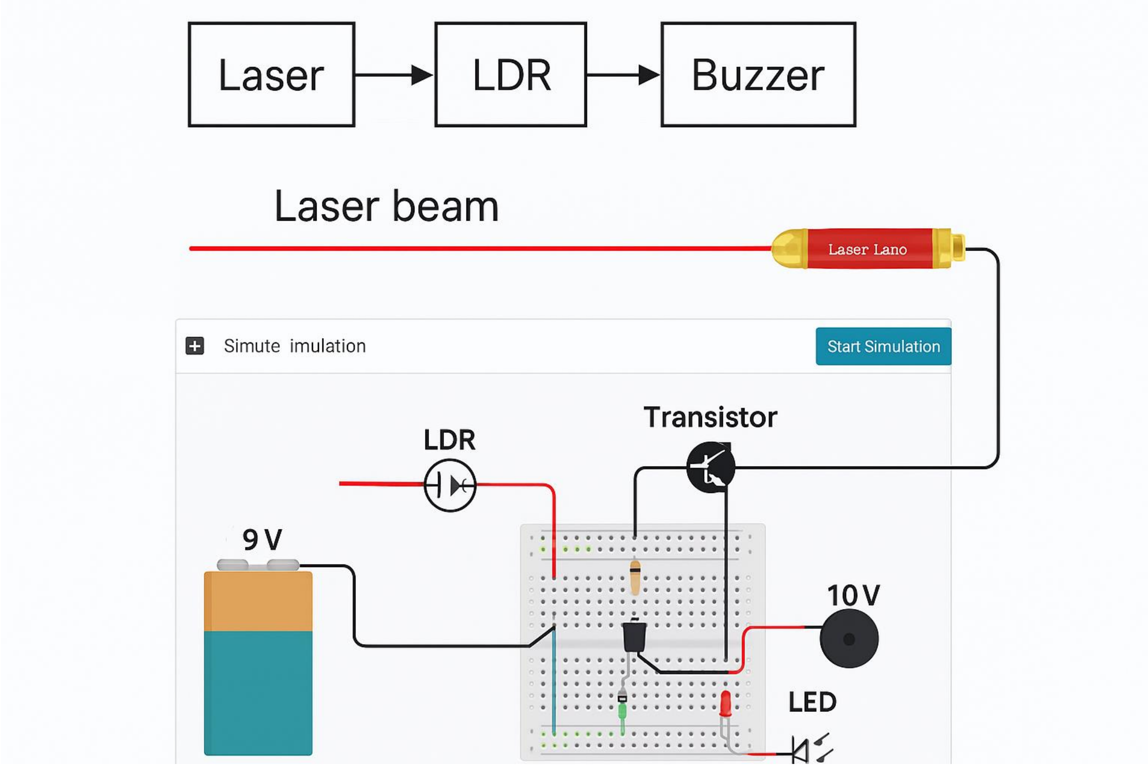
- Photosensitivity of the LDR for detecting light.
- Voltage divider rule for analog signal control.
- Transistor switching for digital control of output devices like buzzers.

Theory and Methodology

- Laser shines on LDR to keep resistance low.
- Beam interrupted → LDR resistance increases → transistor activates buzzer.

Concepts used:

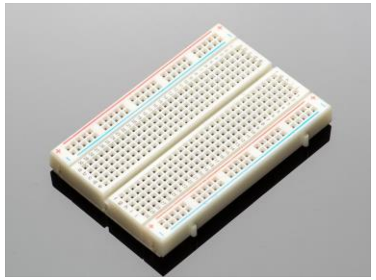
1. LDR (Light Dependent Resistor)
2. Transistor (BC547) switching
3. Voltage divider



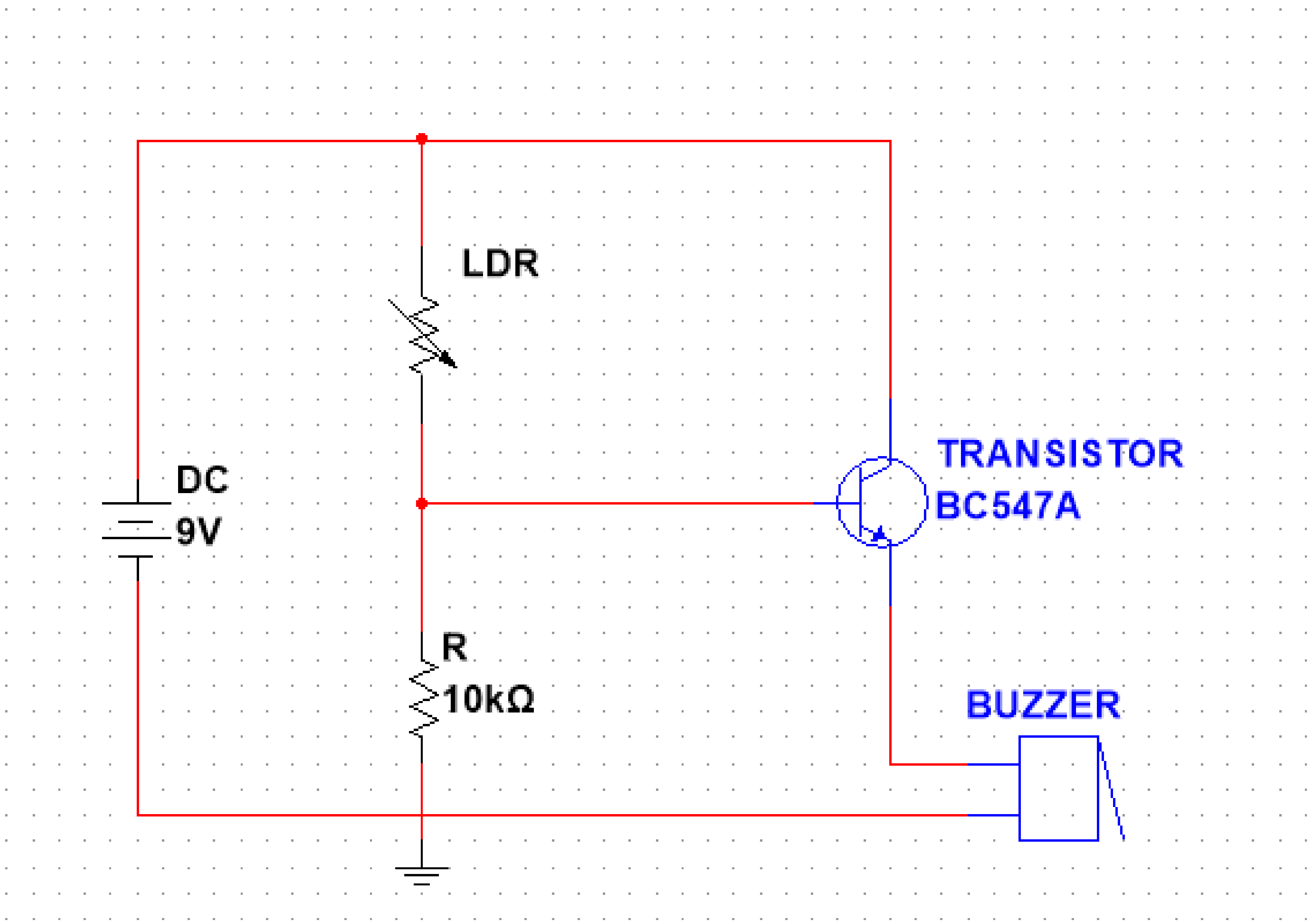
Process of Laser Security Alarm

Equipment

1. 9V DC Battery
2. Breadboard
3. Laser Light
4. LDR
5. 10kΩ Resistor
6. BC547 NPN Transistor
7. Buzzer
8. LED
9. Connecting wires




Circuit Diagram



Procedure

Power Source

- 9V battery connected via clip to breadboard

Voltage Divider Setup

- LDR + 10kΩ resistor in voltage divider
- LDR → 9V DC
- LDR-Resistor junction → BC547 base
- Resistor → GND

Transistor Connection

- BC547 emitter → GND
- Collector → Buzzer (one end)
- Buzzer other end → +9V

Operation

- Laser aimed at LDR (constant light)
- Interruption of beam → Buzzer activates

Data and Analysis

CONDITION	LDR RESISTANCE	V_{BASE} (APPROX)	TRANSISTOR STATE	BUZZER
Laser on LDR	~ 1kΩ	~ 8.2V	OFF	Silent
Laser interrupted	~ 100kΩ	~ 0.V	ON	ON

LDR Resistance Changes

Voltage Divider Formula,

$$V_{out} = V_{in} \times \frac{R2}{R1+R2}$$

Here,

$$R_1 = \text{LDR},\ R_2 = 10k\Omega,\ V_{in} = 9V$$

Example Calculation (Laser Block),

$$V_{out} = 9 \times \frac{10k}{100k+10k}$$

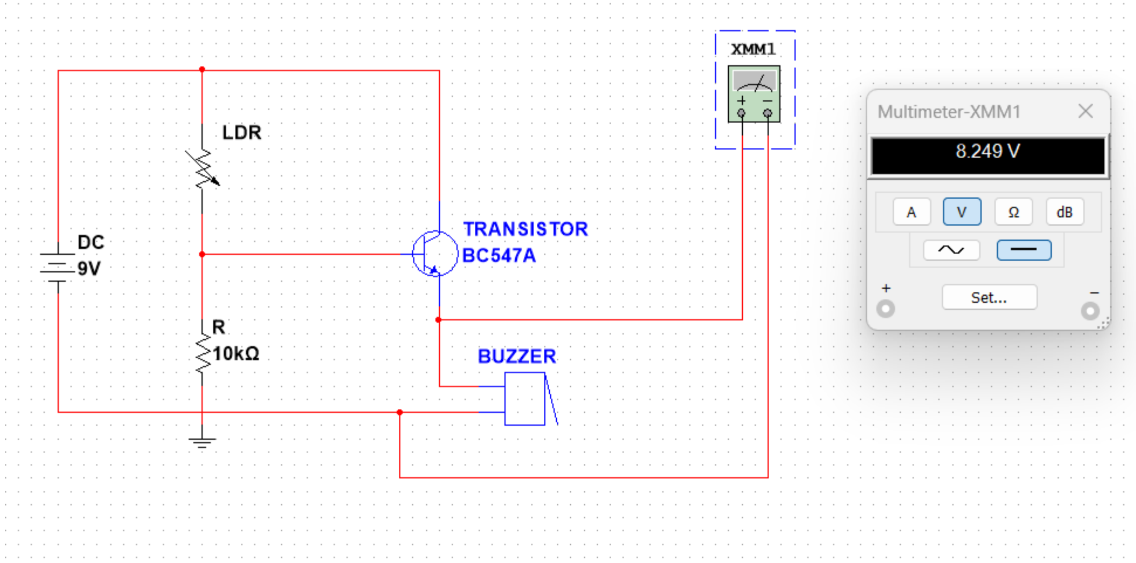
$$= 9 \times 0.090982 \text{ V}$$

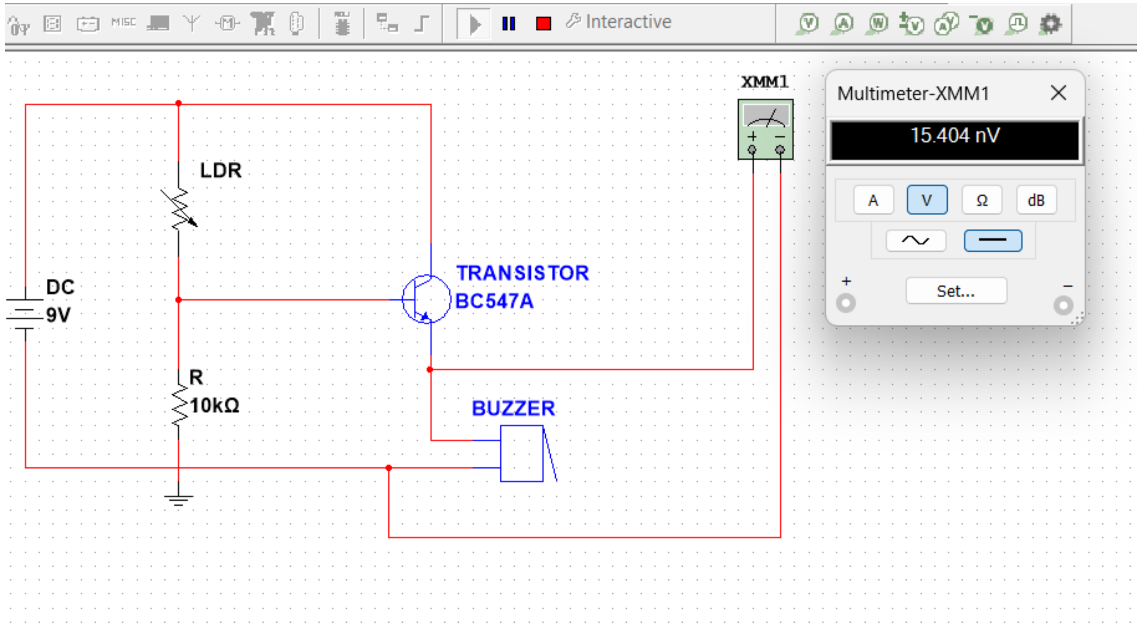
$$= 0.7 \text{ V}$$

When $V_{base} < 0.7V$, the transistor turns ON, and the buzzer Sounds.

Simulation

LDR's Resistance increases→ Turn on Transistor Switch→ Buzzer Active





LDR is exposed to the laser, its resistance is low

Discussion

Pros:

Simple, sensitive, low cost

Cons:

Affected by ambient light, laser alignment, not weatherproof

Improvements:

Light shielding, comparator IC

Uses:

Security systems, Arduino-based monitoring, area coverage with multiple setups

Conclusion

- Effective low-cost security solution.
- Demonstrates sensor-based control.
- Useful in real-world alarm systems.
- Enhances understanding of transistor switching.

THANK YOU!

ANY QUESTIONS?